Interpretability

AI506: Advanced Machine Learning

Spring 2026

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Keywords

- Interpretability and explanations
- Human vs. functional groundedness
- Intrinsic vs. post-hoc interpretability
- Causal interpretability
- Mechanistic interpretability

Why is interpretability important?

Ethics: Aligned with common moral values?

AI Safety: Does the model perform within expectations?

Accountability: When does the model fail in production? (cf. EU AI Act)

Scientific Understanding: Generate hypothesis and knowledge

What is interpretability?

- *Interpretability* is the “ability to explain or to present in understandable terms to a human” (Doshi-Velez & Kim, 2017)
- Interpretability is the degree to which a human can understand the cause of a decision. (Miller, 2019)
- Effective explanations must be selective [ideally only 1-2 causes] (Miller, 2019)

What makes a good explanation?

- **Application-grounded**: Explanations are evaluated in the environment where the model is deployed (ideal but usually not feasible).
- **Human-grounded**: Is the explanation useful to humans?
- **Functionally-grounded**: Does the explanation actually have something to do with what happens in the model?

Advanced machine learning finds stuff and decides things

- But **how** does it finds things and **why** does it make certain decisions?
- Deep nets are extremely powerful, but very poorly interpretable.
- What can we do to make them more interpretable?

Intrinsic vs. post-hoc interpretability

Intrinsic = interpretability integrated in the model

Post-hoc = interpret the model after it has been trained

Feature attribution

The goal is to find explanations that tell us what input features matter for the model's output.

Let's look at a few methods!

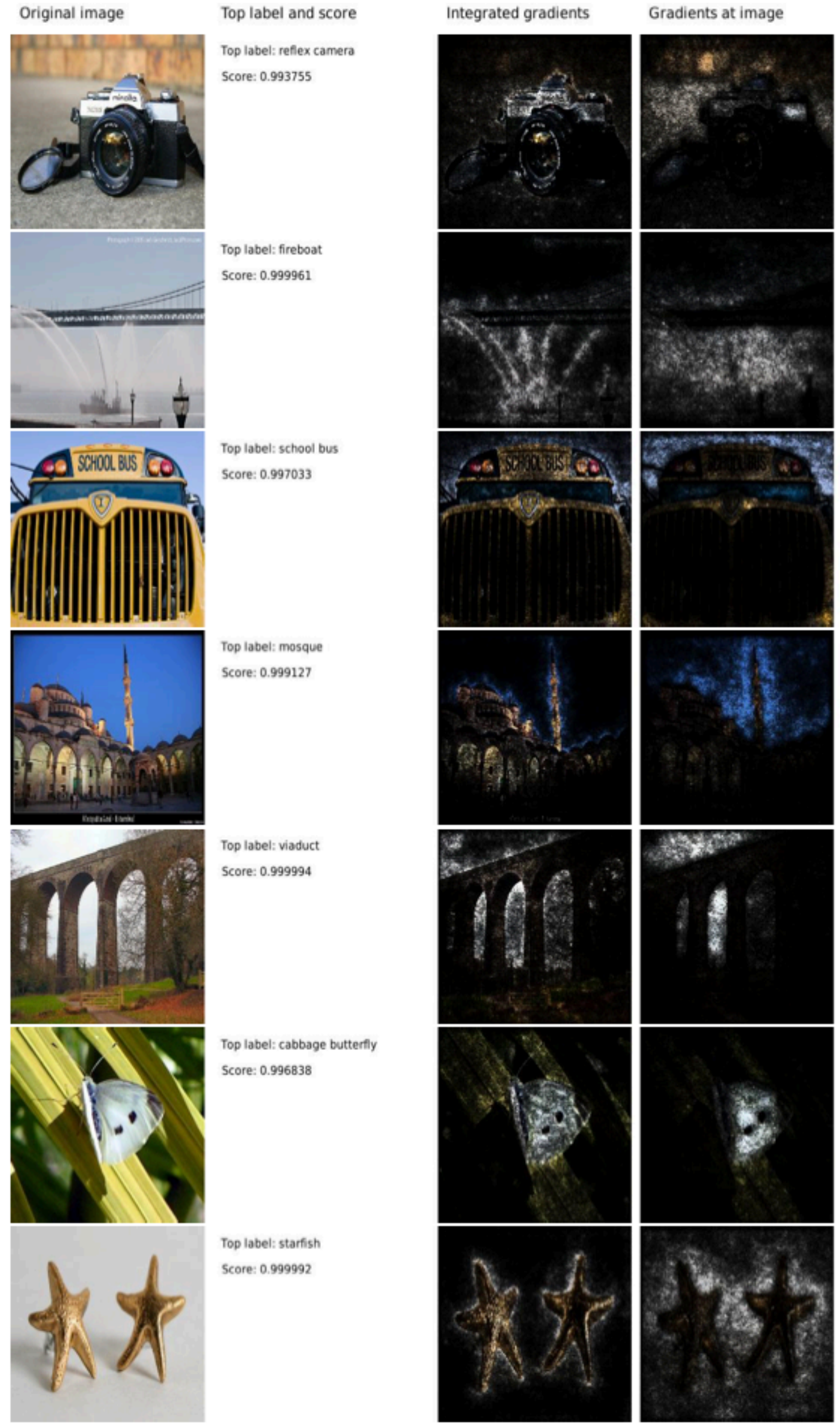
Common post-hoc methods

LIME: Train a linear regression (interpretable!) to predict the output of the studied model.

SHAP: Test all possible combinations of inputs, and observe when the model output changes.

Axiomatic post hoc methods

Integrated gradients: Calculate the derivative with respect to the input (=quantify the effect a small change in input would have on the output)



Model-specific approaches

Just look at the attention values and take those as feature attribution.

2019

- Sarah Wiegreffe, Yuval Pinter:
  Attention is not not Explanation. EMNLP/IJCNLP (1) 2019: 11-20
- Sarthak Jain, Byron C. Wallace:
  Attention is not Explanation. NAACL-HLT (1) 2019: 3543-3556

There are many more

less information → more information

	post-hoc: black-box	dataset	gradient	embeddings	white-box	intrinsic: model specific
local explanation (lower abstraction)						
input features	SHAP § 6.4	LIME § 6.3, Anchors § 6.5	Gradient § 6.1, IG § 6.2			Attention
adversarial examples	SEA$^{\mathcal{M}}$ § 7.2		HotFlip § 7.1			
influential examples		Influence Functions$^{\mathcal{H}}$ § 8.1 TracIn$^{\mathcal{C}}$ § 8.3		Representer Pointers† § 8.2		Prototype Networks
counter-factuals	Polyjuice$^{\mathcal{M},\mathcal{D}}$ § 9.1	MiCE$^{\mathcal{M}}$ § 9.2				
natural language	CAGE$^{\mathcal{M},\mathcal{D}}$ § 10.1					GEF$^{\mathcal{D}}$, NILE$^{\mathcal{D}}$
class explanation						
concepts					NIE$^{\mathcal{D}}$ § 11.1	
global explanation						
vocabulary				Project § 12.1, Rotate § 12.2		
ensemble	SP-LIME § 13.1					
linguistic information	Behavioral Probes$^{\mathcal{D}}$ § 14.1			Structural Probes$^{\mathcal{D}}$ § 14.2	Structural Probes$^{\mathcal{D}}$ § 14.2	Auxiliary Task$^{\mathcal{D}}$
rules (higher abstraction)	SEAR$^{\mathcal{M}}$ § 15.1	Compositional Explanations of Neurons† § 15.2				

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- Probes
- Causal Interpretability
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Linear Probes

A (linear) probe is a (linear) classifier that takes the activations of another models as input and predicts a property of interest.

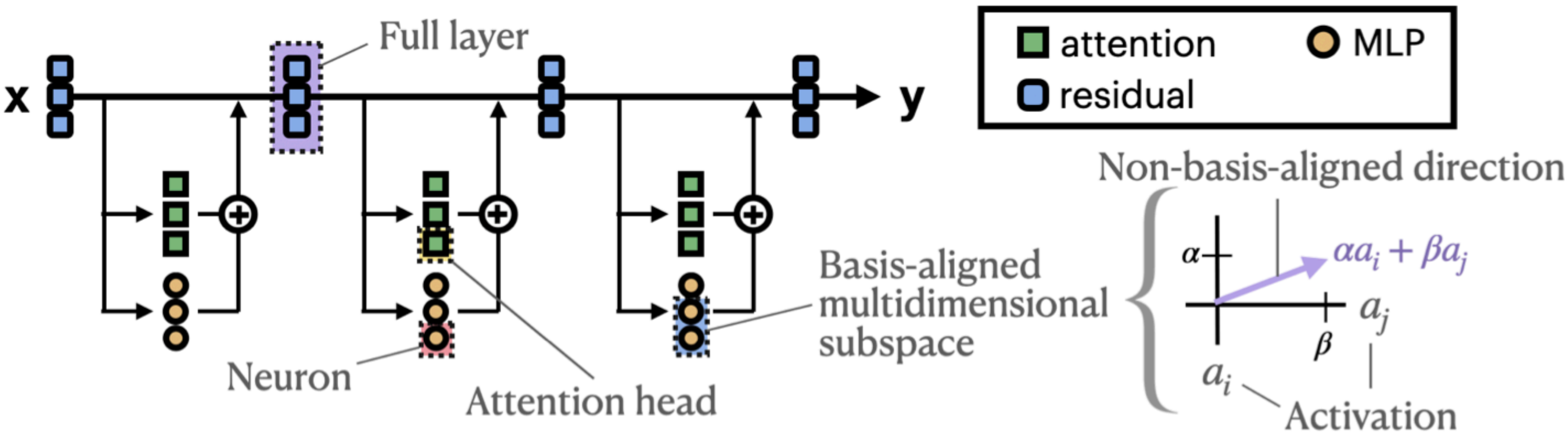
This tells us whether the property of interest is linearly separable in the representation.

In practice: Keep some held-out data, to evaluate how well the probe generalizes.

Causal Interpretability

- Based on Causal Mediation Analysis (Pearl): Does X cause Y through some intermediate variable Z?
  Think of the z as an intermediate hidden layer representation of a neural net

But what is the right mediator?



Open question!

Mechanistic Interpretability

Reverse engineering neural networks

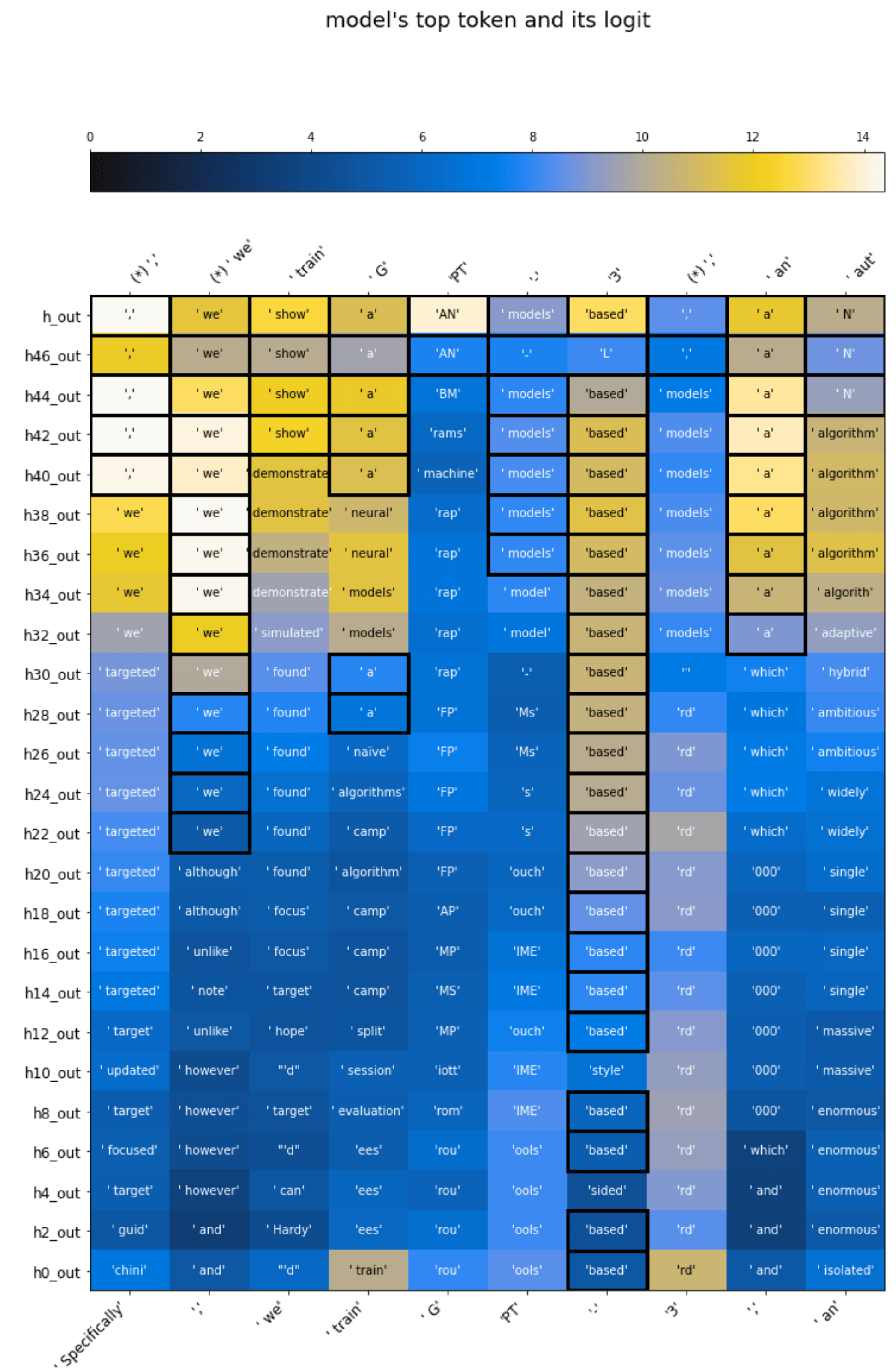
Identifying features and circuits

Causal analysis ("mechanisms")

Mech Interp goes beyond input features and seeks to figure out what mechanisms in the model leads to the output.

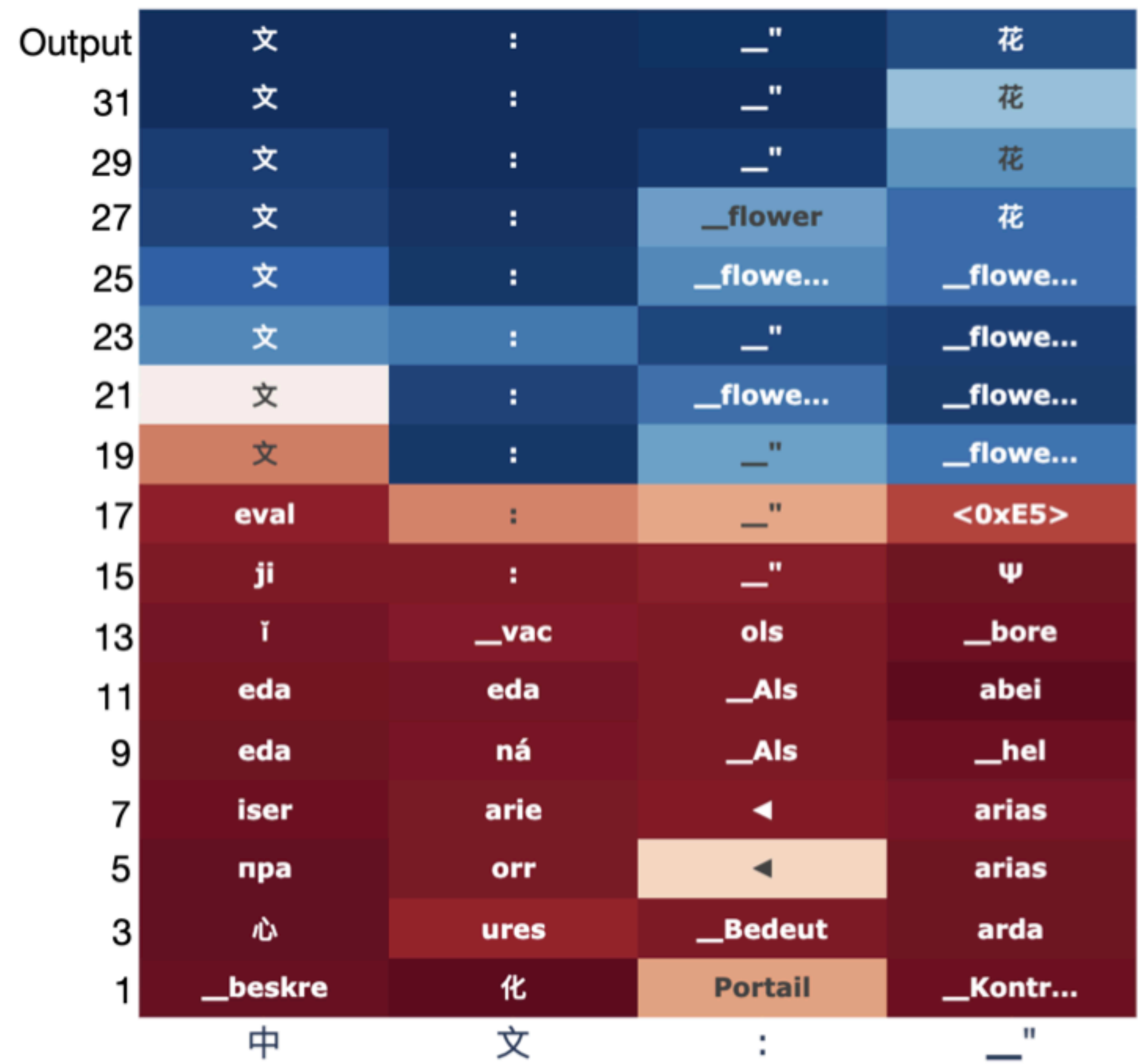
The logit lens

Prematurely apply the final language modeling head to intermediate representations of a transformer



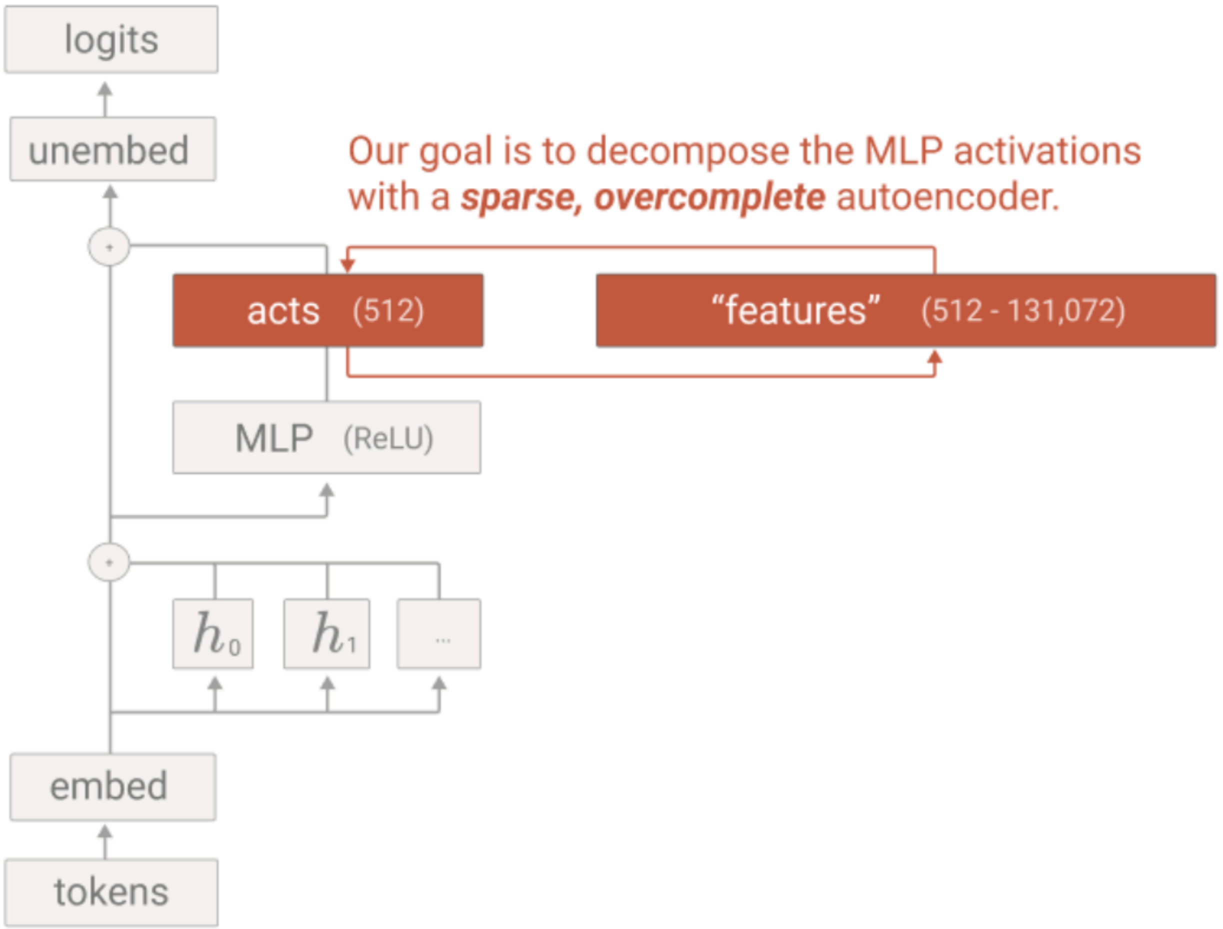
Do Llamas work in English?

Multilingual language models "think" in a concept space that happens to be closest to English (dominant training language)

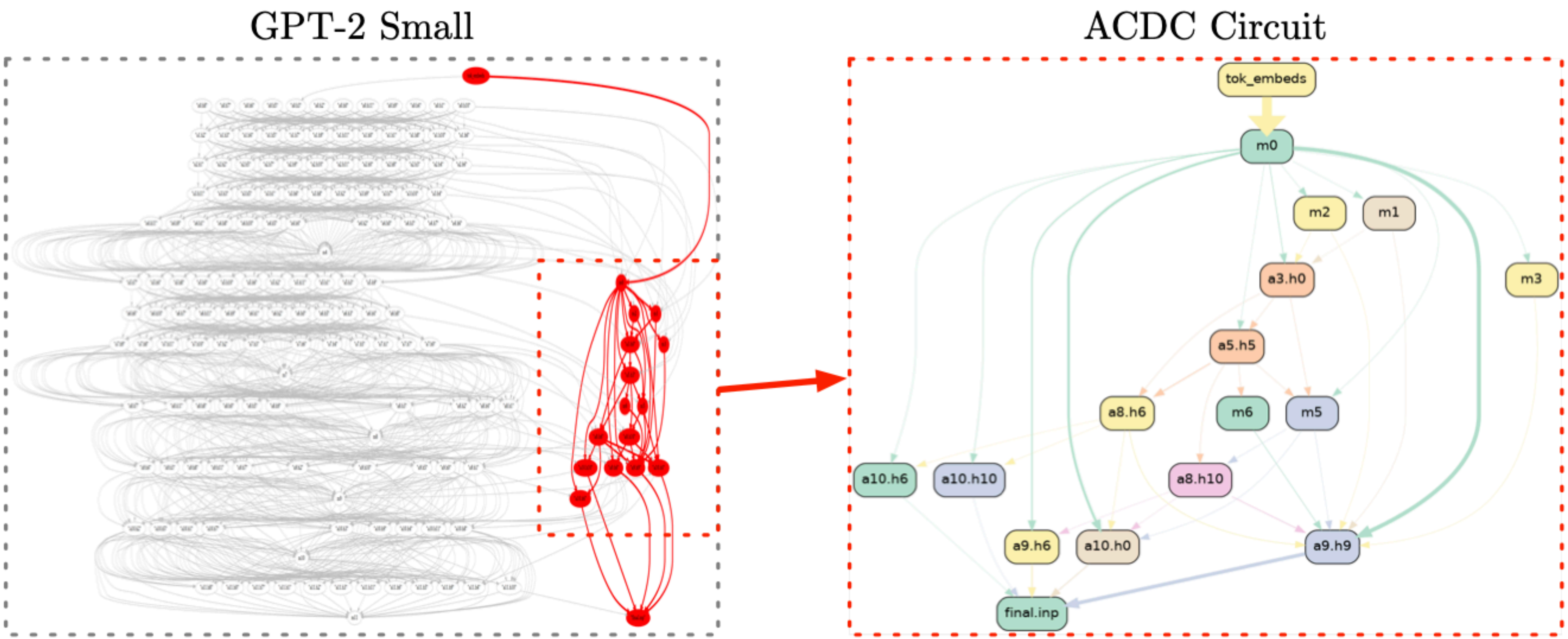


Sparse Autoencoders

Train a sparse autoencoder on the activations of another neural network to learn meaningful features.

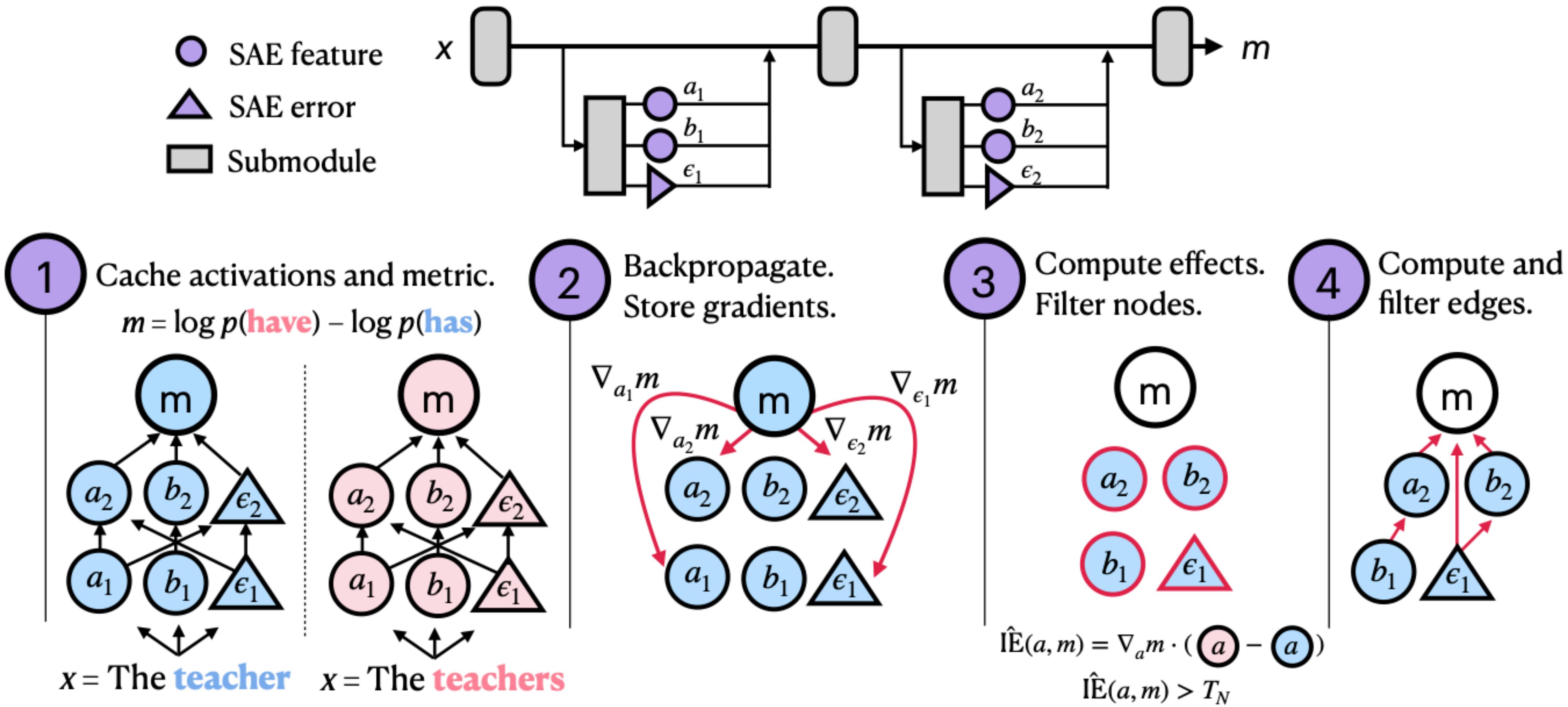


Automated Circuit Discovery



Sparse feature circuits

SAEs + Circuit Discovery Pipeline



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How's my teaching?

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